Autonomous Mobile Robots

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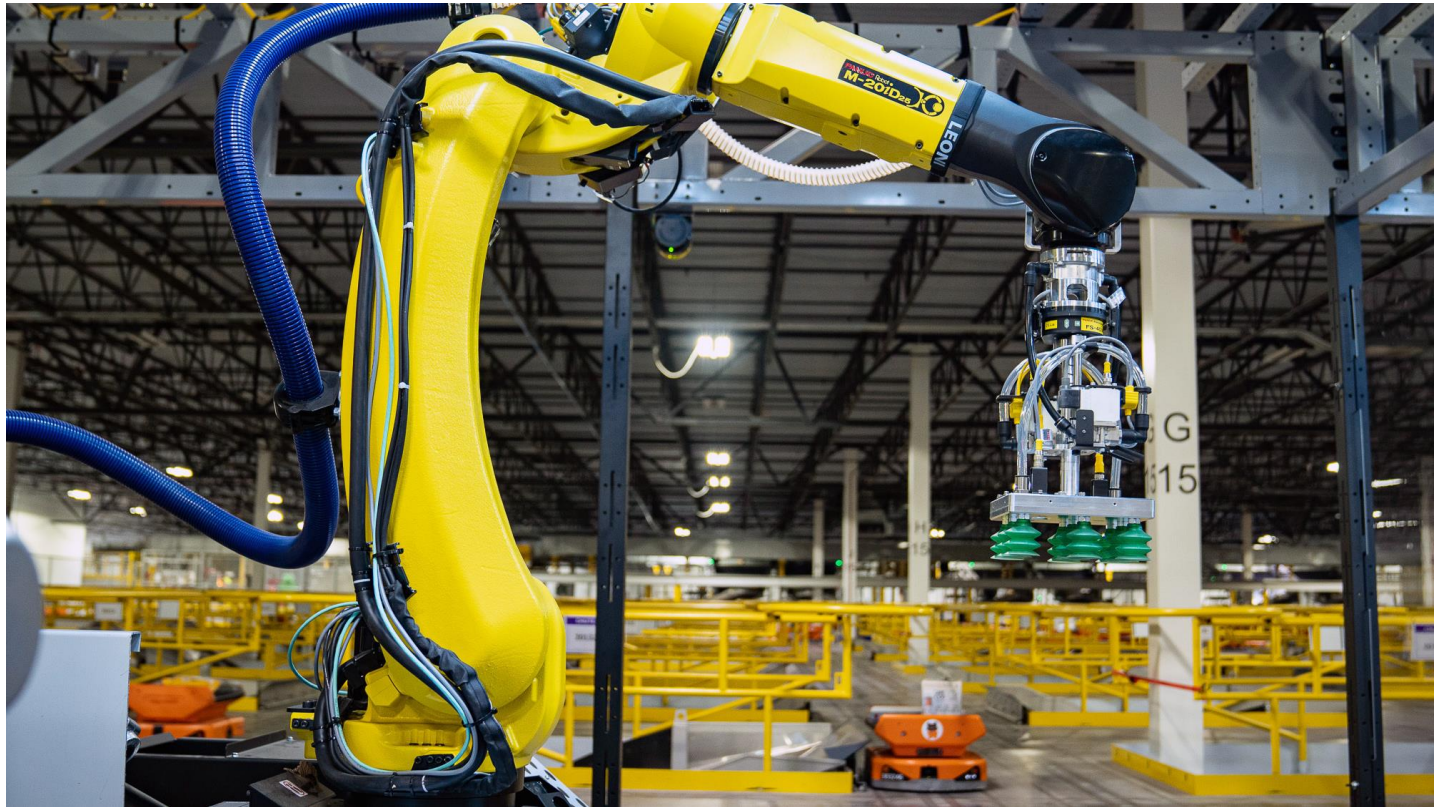
What are Autonomous Mobile Robots?

Intelligent, self-navigating robots that are designed to move materials safely through dynamic environments.



How do AMRs operate?

- Equipped with advanced sensors to detect surroundings
- Process data instantly
- Navigate independently without fixed tracks
- Work safely alongside humans

Industrial Impact

- Transition from manual labour in factories to automated robots.
- Manufacturers prefer AMRs to run at peak efficiency and accuracy.
- AMRs reduce labour costs by 40% and shortens construction time by 25%.
- Business are going from rigid, manual operations to flexible, efficient solutions.

Pros

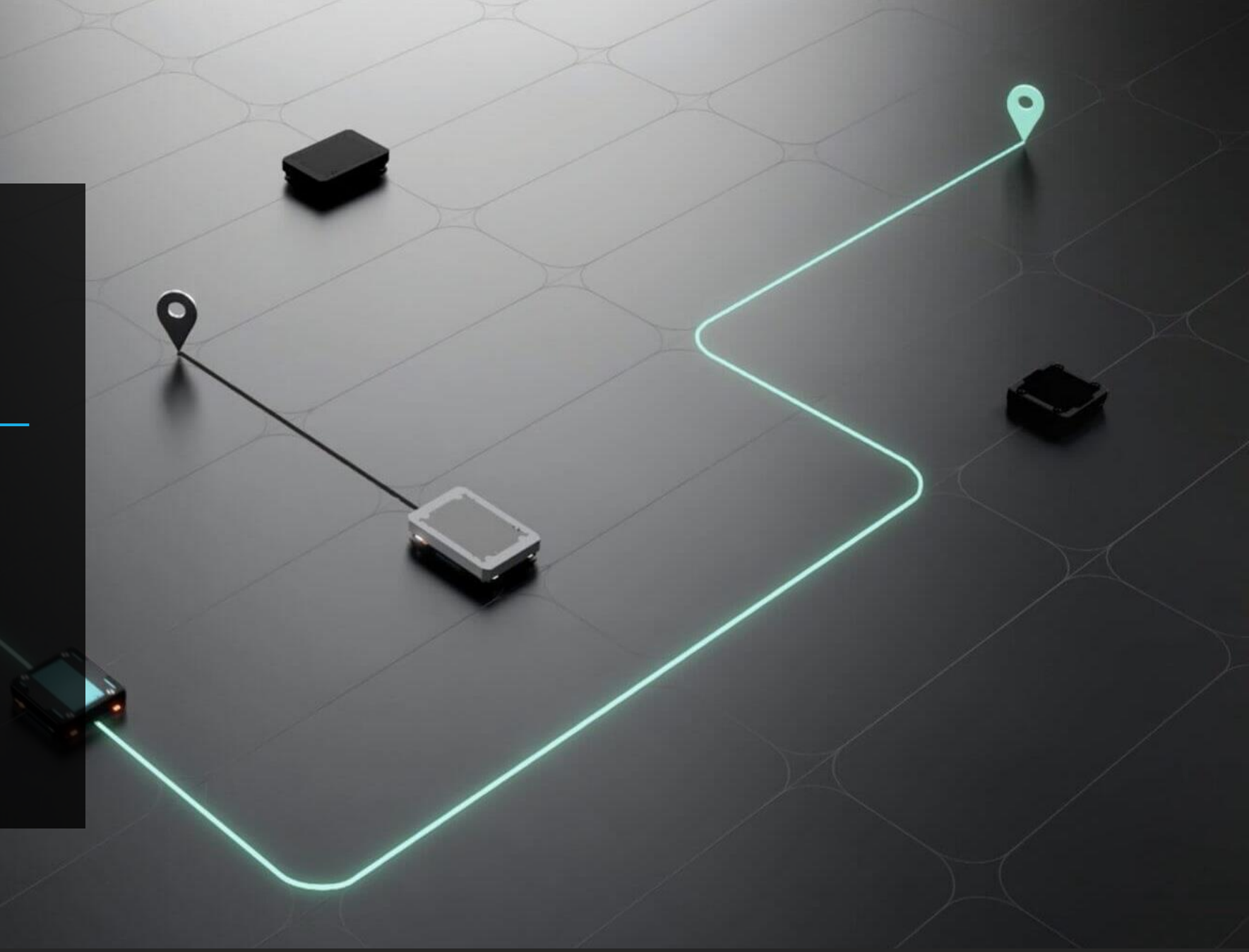
Adaptable to any environment

Navigates best routes

Takes on repetitive tasks

Ensures safety standards

Better long-term investment



Cons

- Implementation requires more infrastructure
- Requires good Wifi and bandwidth (preferably 5G)
- Integration costs are high
- Moves slowly and needs charging.
- Interoperability issues for customers

Business perspective

‘Robotics-as-a-Service’ gaining a competitive advantage

Small businesses are utilizing ‘Robotics-as-a-Service’

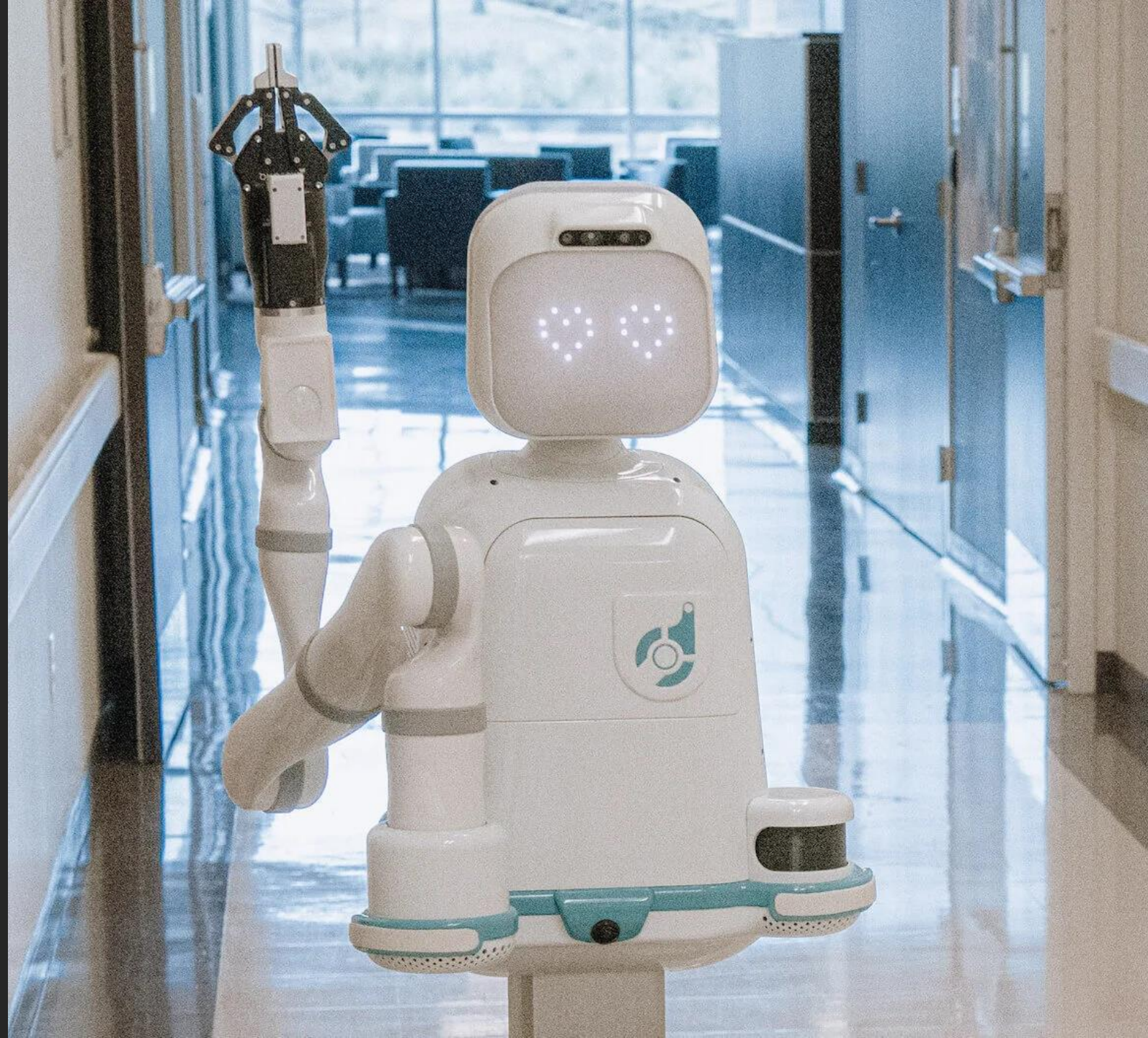
Companies are recognizing the value of AMRs.

Oz by Naïo Technologies

- This AMR autonomously weeds rows of crops without damaging plants using precise row navigation.
- It reduces the need for chemical herbicides
- It is commonly used in small farms, primarily for vegetable crops.

Moxi by Diligent Robotics

- This AMR transports medication, lab samples, and supplies efficiently in hospitals.
- It uses a robotic arm and sensors for navigation, interaction, and delivery.
- It reduces nurse workload, allowing staff to prioritize direct patient care and responsibilities



Whiz by SoftBank Robotics

- This AMR uses AI-powered navigation to map sources and clean floors autonomously.
- Detects obstacles and people, ensuring safe operation alongside humans in busy environments
- It is commonly used in airports, hotels and offices.

Future of AMRs

Emerging Capabilities and Applications

- AI and machine learning will enable greater autonomy and real-time decision making
- Expansion beyond transportation and into monitoring, data collection, and decision support
- Increased interconnectedness between robots and enterprise systems
- Greater reliance by businesses to improve efficiency, accuracy, and flexibility
- Research highlights that:
  "Advances in artificial intelligence and machine learning are powering the next generation of autonomous robots, which will enable them to quickly adapt to new, harsh conditions and make real-time decisions" *(Konieczna-Fuławka et al., 2025*, p. 12.*)*

Summary

AMRs are technological advances that will revolutionize industries reliant on manual labour

Replacing physically straining tasks with automation will boost productivity and efficiency rates

The impact is being recognized in business, as it boosts sales and reduces processing time

The Agricultural, Cleaning, and Hospital AMRs demonstrate the industries where this technology is currently implemented.

In the future, AI will help control AMRs, transforming them into something greater and more beneficial toward humanity

References

Konieczna-Fuławka, M., Koval, A., Nikolakopoulos, G., Fumagalli, M., Santas Moreu, L., Vigara-Puche, V., Müller, J., & Prenner, M. (2025). *Autonomous mobile inspection robots in deep underground mining—The current state of the art and future perspectives*. **Sensors, 25**, 3598. https://doi.org/10.3390/s25123598